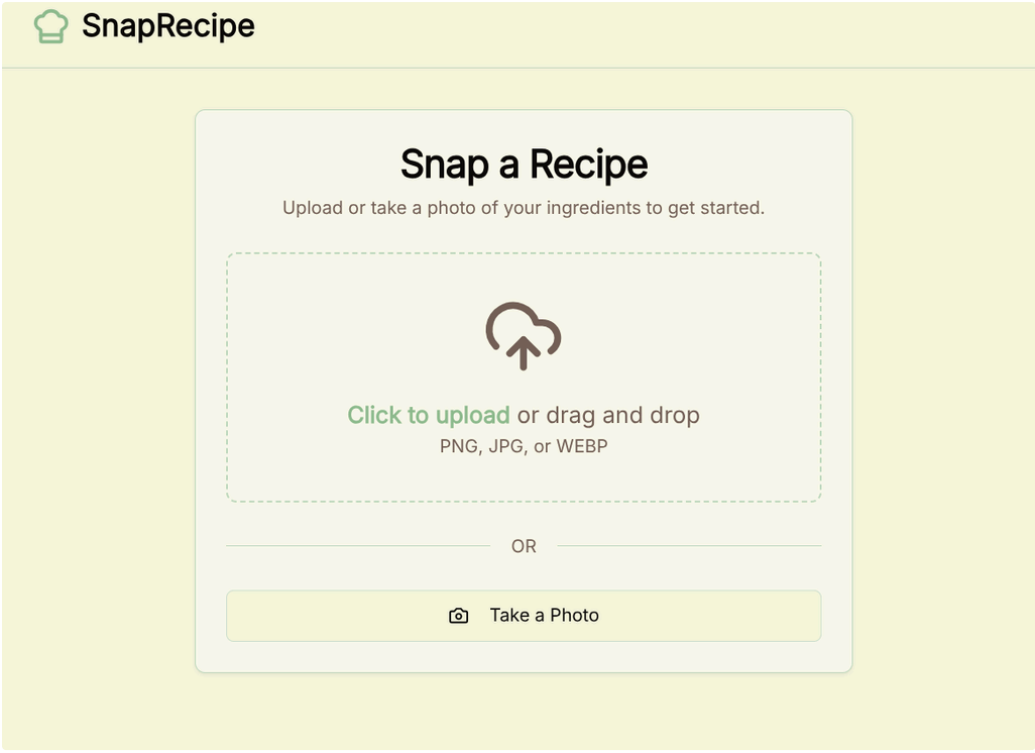
[PRD] SnapRecipe

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1. Introduction

This document outlines the requirements for "SnapRecipe," a smart kitchen application designed to simplify meal preparation by generating recipes from photos of available ingredients. The application will leverage artificial intelligence to identify ingredients in user-uploaded or captured images and then provide relevant, detailed cooking instructions. The primary goal is to offer a seamless and intuitive experience for users looking for quick meal inspiration and efficient use of their pantry items.



2. Goals

- Primary
  - To enable users to effortlessly generate viable recipes by simply taking or uploading a photo of their ingredients.
  - To accurately identify various food ingredients from images using AI.
  - To provide clear, step-by-step recipe instructions based on the detected ingredients.
- Secondary

- To inspire users with new cooking ideas and expand their culinary repertoire.
- To reduce food waste by encouraging the use of existing ingredients.
- To create an engaging and user-friendly platform for home cooks.

3. Target Audience

- Primary
  - Home cooks who frequently look for recipes based on what they have on hand.
  - Individuals seeking quick and convenient meal solutions.
  - Users interested in leveraging technology to simplify their cooking process.
- Secondary
  - Environmentally conscious individuals aiming to reduce food waste.
  - Beginners in cooking looking for guided recipes.
  - Users who enjoy experimenting with different cuisines.

4. Assumptions

- Users have access to a stable internet connection.
- Users have a device capable of taking and uploading high-quality photos (smartphone, tablet, or desktop with camera/upload capability).
- The AI ingredient recognition model will be able to identify a wide range of common kitchen ingredients with reasonable accuracy.
- Users understand basic cooking terminology and techniques.
- The generated recipes will be practical and feasible for home cooking.

5. User Stories

- **US-001**: As a user, I want to be able to upload a photo of my ingredients so that the app can identify them.
- **US-002**: As a user, I want to be able to take a photo directly within the app so that I can quickly capture my ingredients.
- **US-003:** As a user, I want the app to accurately recognize multiple ingredients from my photo so that I get relevant recipe suggestions.
- **US-004:** As a user, I want to see a list of detected ingredients after the image is processed so that I can confirm their accuracy.
- **US-005:** As a user, I want the app to generate recipes based on the ingredients it recognizes so that I can easily find something to cook.
- **US-006:** As a user, I want to view detailed, step-by-step instructions for a selected recipe so that I can cook it effectively.
- **US-007:** As a user, I want to know what image formats are supported for upload (e.g., PNG, JPG, WEBP).
- **US-008:** As a user, I want a clear call to action to upload or take a photo when I first open the app.

6. Product Features

6.1 UI & UX

- **Home Screen:**

- Clean, minimalist design with "SnapRecipe" logo prominently displayed.
- Centralized "Snap a Recipe" section with clear instructions: "Upload or take a photo of your ingredients to get started."
- Prominent upload area with "Click to upload or drag and drop" functionality.
- Supported file formats (PNG, JPG, WEBP) clearly stated below the upload area.
- Visually distinct "OR" separator.
- Clearly labeled "Take a Photo" button for direct camera access.

- **Ingredient Recognition Screen (Post-Upload/Capture):**
  - Loading indicator while AI processes the image.
  - Display of recognized ingredients, possibly with options to confirm or correct.
- **Recipe Display Screen:**
  - List of generated recipes based on detected ingredients.
  - Each recipe entry should include a name, and potentially a small image or key ingredients.
  - Upon selecting a recipe, detailed instructions should be presented in a clear, readable format.
  - Navigation to go back to ingredient input or recipe list.

6.2 Functional Requirements

- **FR.1 Image Input:**
  - FR.1.1 Users must be able to upload image files (PNG, JPG, WEBP) from their device.
  - FR.1.2 Users must be able to directly capture an image using their device's camera within the app.
- **FR.2 AI Ingredient Recognition:**
  - FR.2.1 The system shall employ an AI model to identify food ingredients present in the uploaded or captured image.
  - FR.2.2 The system shall be able to identify multiple distinct ingredients in a single image.
  - FR.2.3 The system shall display a list of recognized ingredients to the user for confirmation.
  - FR.2.4 The system should handle cases where ingredients are not recognized or are recognized incorrectly (e.g., suggest alternatives, allow manual input).
- **FR.3 Recipe Generation:**
  - FR.3.1 Based on the confirmed list of ingredients, the system shall generate a list of suitable recipes.
  - FR.3.2 Recipes should be generated leveraging a comprehensive recipe database.
  - FR.3.3 The system should prioritize recipes that primarily use the detected ingredients, minimizing the need for additional items.
- **FR.4 Recipe Display:**
  - FR.4.1 For each generated recipe, the system shall display a clear recipe name and an overview.
  - FR.4.2 Upon selection, the system shall display detailed, step-by-step cooking instructions.
  - FR.4.3 Recipe details should include ingredients list, preparation time, cooking time, and serving size.
- **FR.5 Error Handling:**
  - FR.5.1 The system shall provide clear error messages for invalid file types or upload failures.
  - FR.5.2 The system shall inform the user if no recipes can be generated from the detected ingredients.

6.3 Data Requirements

- **DR.1 User Data:**
  - User-uploaded/captured images (temporarily stored for processing).
  - Session-specific ingredient lists.
- **DR.2 Ingredient Database:**
  - Comprehensive list of recognizable ingredients.
  - Metadata for each ingredient (e.g., common synonyms, categories).
- **DR.3 Recipe Database:**
  - Extensive collection of recipes.
  - Each recipe should include: Name, description, ingredients list (with quantities), step-by-step instructions, preparation time, cooking time, serving size, and potentially dietary tags.
  - Relationships between recipes and ingredients.
- **DR.4 AI Model Data:**
  - Training data for ingredient recognition.
  - Model weights and configurations.

7. Non-Functional Requirements (Performance, Usability, Reliability, Security)

- **Performance:**
  - NFR.1.1 Image upload/capture to ingredient recognition: Under 5 seconds for typical images.
  - NFR.1.2 Ingredient recognition to recipe generation: Under 5 seconds.
  - NFR.1.3 Recipe display loading time: Under 2 seconds.
  - NFR.1.4 The application should remain responsive under peak usage.
- **Usability:**
  - NFR.2.1 The user interface must be intuitive and easy to navigate for first-time users.
  - NFR.2.2 All text and instructions must be clear, concise, and easy to understand.
  - NFR.2.3 The design must be responsive and accessible across various device screen sizes (mobile, tablet, desktop).
- **Reliability:**
  - NFR.3.1 AI ingredient recognition accuracy rate: Target 85% for common ingredients.
  - NFR.3.2 Recipe generation engine uptime: 99.9% availability.
  - NFR.3.3 The application should gracefully handle network interruptions and API failures.
- **Security:**
  - NFR.4.1 User image data must be handled securely and temporarily, with clear data retention policies.
  - NFR.4.2 All data transmission between the client and server must be encrypted (e.g., HTTPS).
  - NFR.4.3 User privacy must be maintained in accordance with relevant data protection regulations.

8. Release Criteria

- All core features (image input, AI recognition, recipe generation, recipe display) are implemented and functional.
- User Acceptance Testing (UAT) is completed with positive feedback from a representative group of target users.
- All critical bugs are resolved.

- Performance metrics meet or exceed defined targets.
- Security audit completed with no high-severity vulnerabilities found.
- Comprehensive test coverage achieved (unit, integration, end-to-end tests).
- Documentation (user guides, API documentation for internal use) is complete.

9. Success Metrics

- **SM.1 User Engagement:**
  - Number of daily/weekly active users (DAU/WAU).
  - Average number of recipes generated per user per week.
  - Retention rate (e.g., percentage of users who return after 7 days).
- **SM.2 Feature Usage:**
  - Percentage of users utilizing the "Take a Photo" vs. "Upload" feature.
  - Number of recipes viewed in detail.
- **SM.3 Recognition Accuracy:**
  - Manual review of AI recognition accuracy for a sample of images.
  - User feedback on incorrect ingredient detection.
- **SM.4 User Satisfaction:**
  - Net Promoter Score (NPS) or user satisfaction surveys.
  - App store ratings and reviews.